

EFS01: Lecture Summary

Trading System: A complete trading system includes:

- **Entry Point**
 - What to Buy
 - When to Buy
 - How much to Buy (Position Management)
- **Exit Point**
 - What price or what time to take profit
 - Stop Loss

Moving Average Crossover System:

- Hypothesis: A sustainable and profitable trend can be identified by using a crossover of smaller moving average and larger moving average.
- Back-testing: It should be conducted in-sample and out-of-sample
- Testing Period should incorporate a wide range of markets:
 - Bullish phase
 - Bearish phase
 - Consolidation phase
 - Distribution phase
- Analyse:
 - The Results on various Parameter values and optimisation.
 - Distribution of Losses
 - Position Sizing
- Stop-loss and Take-profit conditions

P&L Analysis for Trading System:

- Total Profit: Simple summation of profits(Assuming we are dealing with daily simple returns of a stock, which are almost identical to daily log returns and hence can be summed up to get the total return)
- Hit Ratio-Success Ratio: Probability of a trade being profitable
- Average Profit/Loss Per Trade: (Sum of Profits)/Total Profitable Trades
- Loss Per Trade: (Sum of Loss)/Total Losing Trades
- Maximum Drawdown: Maximum money lost from a peak portfolio value.
- Variability of returns: Sharpe ratio
- Distribution of profits/losses
- Total Number of Trades

- Expectancy = (Probability-win* Amt-win) –(Probability-loss*Amt-loss)

Costs to be considered in P&L:

- Brokerage: Transaction charges from the broker, brokerages from the exchange and statutory taxes
- Slippage: Slippages can be calculated as the average of bid-ask spread from different volatility periods.

Martingale vs Anti Martingale:

- **Martingale:** You increase your bet size as your account falls, decrease it after a win.
- **Anti-martingale:** You increase your bet size as your account grows, decrease it after a loss

- **Example of Martingale Strategy:**

Type of position/trade:Long

Strategy: Buy on 2% fall and square off in 2% up

BUY		
Price	Units	Amount
100	1	100
98	2	196
96	3	288
94	4	376
92	5	460

Average Buy: 94.66

SELL		
Price	Units	Amount
94	1	94
96	2	192
98	3	294
100	4	400

102	5	510
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Average SELL: 99.33

Note: This is just one of many scenarios of the strategy.

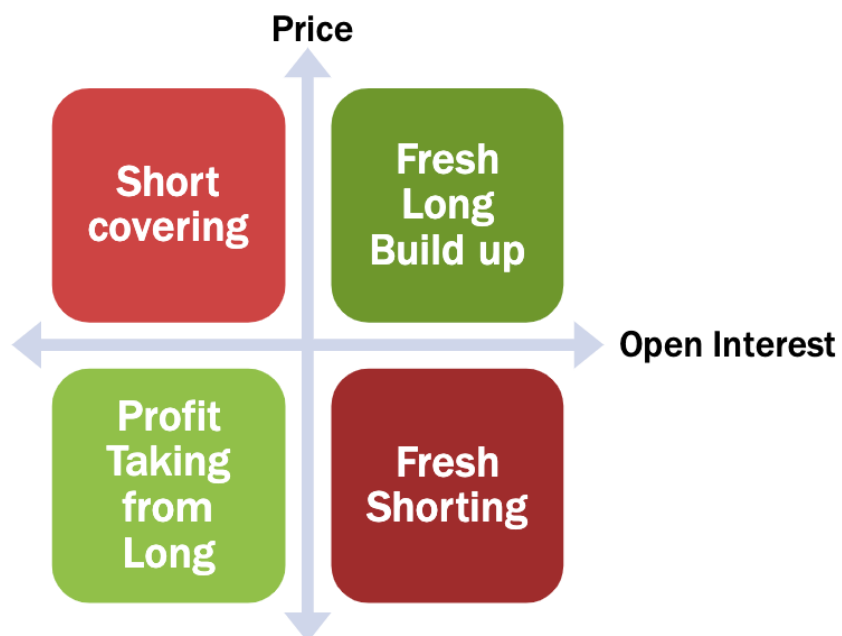
These methods assume dependency between trades.

- In most cases, trades are independent of each other.
- The odds of the next trade being a win are not related to whether the last trade was a win or a loss.

Analysis on Open Interest and Price:

Interpretation:

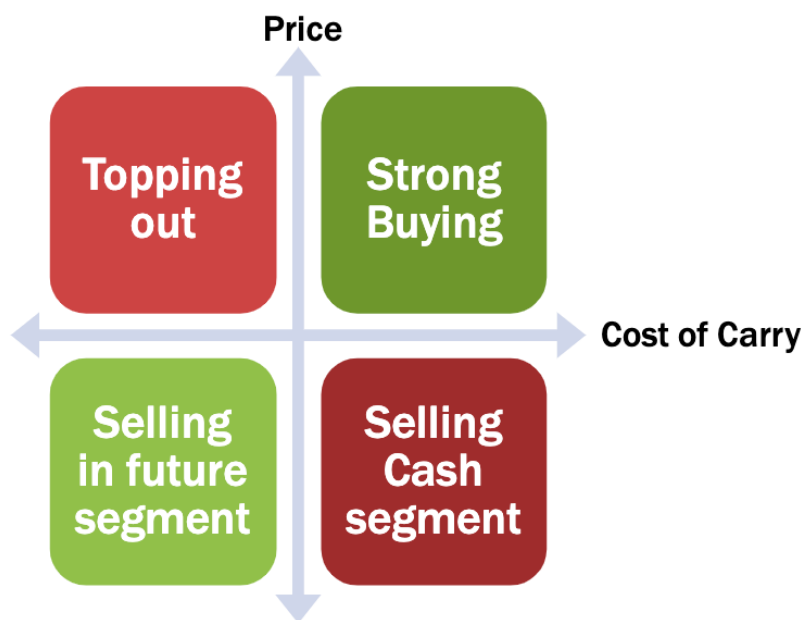
- **Short Covering:** Price goes up and Open Interest goes down
- **Profit Taking from Long:** Price goes down and Open Interest goes down
- **Fresh Long Build up:** Price goes up and Open Interest goes up
- **Fresh Short Build up:** Price goes down and Open Interest goes up



Analysis on “Cost of Carry” and Price:

Interpretation:

- **Topping Out:** Price goes up and COC goes down
- **Selling in the future segment:** Price goes down and COC goes down
- **Strong Buying:** Price goes up and COC goes up
- **Selling Cash segment:** Price goes down and COC goes up



Analysis of Delivery Volume:

- Delivery volume is the volume of stock delivered of a particular script at the end of the trading day.
- Delivery Volume % is the percentage of the volume of the stock delivered with respect to the total traded volume.
- Delivery Volume of the day should be compared with SMA of the same for the given period.
- It can be used in conjunction with OI and Price change
- A weighted system looking at factors like:
 - Change in OI
 - Change in Price
 - Change in Delivery

List of Excel Formulas

- SUM
- VLOOKUP
- AVERAGE
- AND
- ROW
- OFFSET
- MAX

- YEAR
- COUNTIF
- AVERAGEIF
- ABS
- SUMIFS
- SUMIF
- STDEV
- MIN
-

List of Excel Matrix Formulas

- SUMPRODUCT
- FREQUENCY